BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: MRD-IX-21(A)

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-IX-21(A)

Note: Note: There are three sections in this paper i.e., A, B & C. Instructions: Please attempt section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, overwriting or multiple circle shading.

Q1. Write the correct option i.e. A, B, C or D in the empty box provided opposite to each part. Cutting, erasing and over writing will not be awarded any marks.

Q1. The gram molecular mass of HNO_3 is:

(A) 60
(B) 63
(C) 98
(D) 100

Q2. Electron in its ground state does not .....

(A) spin
(B) revolve
(C) radiate energy
(D) reside in orbit

Q3. Which of the following would have the smallest ionization energy?

(A) K
(B) S
(C) P
(D) Ca

Q4. An element x is in group VI of the periodic table. Its ion will be represented by:

(A) X^{+}
(B) X^{-}
(C) X^{+2}
(D) X^{-2}

Q5. At freezing point which one of the following coexists in dynamic equilibrium.

(A) gas and solid
(B) liquid and gas
(C) liquid and solid
(D) all of these

Q6. Which of the following solution is more dilute?

(A) 1.0M (B) 0.5M

(C) 0.05M (D) 0.005M

Q7. What is the oxidation number assigned to manganese in $KMnO_4$?

(A) +3 (B) +2

(C) +7 (D) +4

Q8. The halogen present in the solid form is:

(A) Fluorine (B) Chlorine

(C) Bromine (D) Iodine

Q9. S-block element are:

(A) Metals (B) Non-metals

(C) Metalloids (D) Transition

Q10. The word Alkali means:

(A) Acid (B) Base

(C) Basic salt (D) Ashes

Q11. Which of the following ions do not have the electronic configuration of an argon atom?

(A) S^{-2} (B) Ca^{+2}

(C) O^{-2} (D) K^{+}

Q12. When there is a low concentration of solute in a solution, it is known as:

(A) Dilute solution (B) saturated solution

(C) concentrated solution (D) super saturated solution

Section - B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

QII. Write short answers of any eight (8) of the following parts. All parts carry equal marks.

1. Differentiate between compound and mixture.
2. What is electronic configuration?
3. How will you differentiate between representative and transition elements?
4. Why most atoms are chemically bonded to other atoms in nature?
5. Give an example of a non polar molecule with polar bonds. Give reason.

6. Why are the densities of gases lower than that of liquids?

7. Write factors affecting evaporation.

8. A bottle in a drug store contain a label "3 percent hydrogen peroxide". What does it mean?

9. Why NaOH is a strong but NH_4oH is weak electrolyte?

10. Write down some possible uses of an electrolytic cell.

11. Compare between metals and non metals

Section "C"

Maximum Marks: 21

> Note: Answer any THREE questions. All questions carry equal marks.

Q3. State the postulates of Neil Bohr's atomic theory.

Q4.

a. Distinguish between ionic and covalent bond.

b. Explain how ionization energies of elements vary across a period and a group in the periodic table.

Q5.

a. Define the following terms: (i) Effusion (ii) Condensation (iii) Evaporation

b. Write a note on molarity.

Q6.

(a) Sketch the Daniel cell, labeling the cathode, anode and the direction of flow of the electrons.

(b) What happens during displacement reaction in halogens? Give at least one example.